

## QA Engineering Squad: Progress Metrics

- **Current Progress:** 5% (Commencing Week 1, Day 2 of the 3-Month Bootcamp).
- **Current Focus:** Week 1 Mandate — Edge-Case Generation, Boundary Value Analysis, and Regular Expression (Regex) Validation.

## The Daily Drop Blueprint

**Title:** QA Engineering Squad: Week 1, Tuesday - AI Edge-Case Generation & Regex Data Validation

### Section A: Today's Materials (The Synthesis Mandate)

You must synthesize the information from all sources below to complete today's execution mandate. Do not consult outside tutorials.

#### 1. The Video Drills (Watch These First)

- **Scrimba Anchor 1:** [Introduction to Unit Testing - Module: "Grouping with describe"](#) (*Crucial for learning how to structure and group your test case prerequisites*)[cite: 16].
- **Scrimba Anchor 2:** [Learn Regular Expressions - Module: "Restrict Possible Usernames"](#) (*Mandatory foundation for mathematically validating user inputs*)[cite: 17].
- **External Video 1:** [YouTube: Boundary Value Analysis & Equivalence Partitioning](#) (*Core QA theory for generating negative test cases*).
- **External Video 2:** [YouTube: Regex Tutorial for Beginners](#) (*Programming with Mosh's definitive guide to Regex formatting*).

#### 2. Required Technical Documentation

- **Primary Architecture Doc:** [OWASP Data Validation Cheat Sheet](#) (*The industry benchmark for testing against malicious inputs*).
- **Specific Syntax Doc:** [MDN Web Docs: Regular Expressions](#) (*Required for your data validation logic*).
- **Configuration Doc:** [Jira Official: Test Case Traceability Best Practices](#)

#### 3. Engineering Assets

- **Requirements Asset:** The Product Requirements Document (PRD) provided on Day 1.
- **LLM Prompting Asset:** [OpenAI Official Guide: Prompt Engineering](#) (*Use this to prompt the LLM for the 50 edge-case permutations*).
- **Reporting Asset:** Your live Test Case Matrix (Google Sheets) from Day 1.

## Section B: The Execution Mandate (9:00 AM - 2:00 PM)

**1. Execution Summary (TL;DR) Objective:** Evolve your basic Test Case Matrix by feeding the PRD into an LLM to generate extreme edge-case scenarios, and construct strict Regular Expressions (Regex) to validate frontend inputs against malicious payloads.

## **2. The Workflow:**

- **9:00 AM - 10:00 AM (Learning):** Watch the Regex and Boundary Value drills. Synthesize the MDN documentation on Regex syntax.
- **10:00 AM - 1:00 PM (Building):**
  - **Action 1 (Setup & Grouping):** Open your Test Case Matrix. Utilize the logic from the "Grouping with describe" module to categorize your test cases into strict suites (e.g., Authentication, Input Validation, Checkout)[cite: 16].
  - **Action 2 (AI Edge-Case Generation):** Feed the PRD into ChatGPT. Prompt the LLM to generate 50 extreme edge-case testing scenarios. You must include boundary testing such as Arabic right-to-left strings, 10,000-character inputs, and negative integers in pricing fields.
  - **Action 3 (Regex Construction):** Identify 3 input fields from the PRD (e.g., Email, Username, Phone Number). Generate complex Regular Expressions to automatically validate the data formats for these fields.
- **The Non-Negotiable Rules:**
  1. **The "Happy Path" Ban:** Testing only valid emails and standard names is an automatic failure. You must test malicious creativity, including SQL injection strings and extreme boundary violations.
  2. **The AI Defense (Regex Interrogation):** If you use Copilot or ChatGPT to generate your Regular Expressions, you must be able to mathematically explain the logic. If a Technical Mentor asks what the ^ or \$ symbols do in your regex string and you cannot articulate the answer, your ticket fails and your sprint velocity is penalized.
- **1:00 PM - 1:30 PM (Hygiene):** Verify your edge cases are correctly linked to PRD requirements. Push your updated matrix to the shared QA repository.
- **1:30 PM - 2:00 PM (LinkedIn):** Draft a technical summary of your execution. Detail the concept of Boundary Value Analysis and explain why Regex is a critical tool for Software Test Engineers. You must hyperlink your ProSensia admit card within the post text.



## **Section C: Submission Protocol (Due strictly by 2:00 PM)**

- **Step 1 (The Kanban Board):** Verify that this specific task card is shifted to the "Under Review" column on your team board.

- **Step 2 (The Assignments Module):** Navigate directly to today's entry in the platform's independent Assignment block. Paste both the public link to your updated Test Case Matrix (Google Sheets) and your live LinkedIn Post URL into the designated submission form fields.
- **Operational Note:** All grading and technical feedback loops operate through this portal. Do not DM Management your links[cite: 14].